

QUICK_REFERENCE

HelixCode CLI Agent Integration — Quick Reference Card

SSH Submodule Commands (Run These First)

1. Verify SSH access

```
ssh -T git@github.com
```

2. Add 4 MISSING submodules via SSH (NOT HTTPS)

```
git submodule add --force git@github.com:HelixDevelopment/  
HelixAgent.git submodules/helix_agent
```

```
git submodule add --force git@github.com:HelixDevelopment/  
HelixLLM.git submodules/helix_llm
```

```
git submodule add --force git@github.com:HelixDevelopment/  
HelixMemory.git submodules/helix_memory
```

```
git submodule add --force git@github.com:HelixDevelopment/  
HelixSpecifier.git submodules/helix_specifier
```

3. Initialize ALL submodules recursively

```
git submodule update --init --recursive
```

4. Verify no HTTPS URLs exist

```
grep -n "https://github.com" .gitmodules # Should return NOTHING
```

5. Create Go workspace and build

```
cat > go.work << 'EOF'
```

```
go 1.26
```

```
use (
```

```
·
```

```
· ./HelixCode
```

```
· ./submodules/llms_verifier
```

```
· ./submodules/helix_agent
```

```
· ./submodules/helix_llm
```

```
· ./submodules/helix_memory
```

```
· ./submodules/helix_specifier
```

```
· ./HelixQA
```

```
· ./Challenges
```

```
· ./Containers
```

```
)
```

```
EOF
```

```
go work sync
```

```
go build ./...
```

```
# 6. Commit
git add .gitmodules go.work
git commit -m "feat: Add missing submodules via SSH"
```

Integration Phases at a Glance

Phase	Duration	Key Deliverable	Risk
0 Foundation	5-7 days	All 11 submodules initialized	HIGH
1 Core Infra	14-18 days	LLM gateway + memory + spec	HIGH
2 CLI Foundation	18-22 days	Tool framework + context + edit	MED-HIGH
3 Power Features	21-28 days	Plan mode + sandbox + MCP	HIGH
4 UI/UX	14-18 days	Streaming TUI + themes	MED
5 Testing	14-18 days	100% coverage + QA	MED

Total: ~63 working days

Top 10 Features to Port First

1. **OS-Native Sandboxed Execution** (Codex) — CRITICAL security
2. **Auto-Compaction System** (Claude Code) — Infinite conversations
3. **Permission Rule System** (Claude Code) — 5 modes + wildcards
4. **LSP Integration** (OpenCode) — Semantic code understanding
5. **4-Layer Fuzzy Matching** (Aider) — Robust search/replace
6. **Cumulative Diff Review** (Plandex) — Review ALL before applying ANY
7. **Subagent Delegation** (Kilo Code) — Recursive agent hierarchy
8. **MCP Full Lifecycle** (Claude Code) — 4 transports + OAuth
9. **Plan/Act Dual-Mode** (Cline) — Separate planning from execution
10. **Shadow Git Checkpoints** (Cline) — Granular rollback

Key Files Generated

File	Size	Purpose
stage1_helixcode_mapping.md	43KB	Complete repo tree + 87 submodules
stage1_standalone_repos_mapping.md	7KB	8 standalone repos analyzed
stage2_helixcode_architecture.md	38KB	Internal architecture deep dive
stage2_claude_code_deep_analysis.md	43KB	PRIMARY agent — 20 top features

File	Size	Purpose
stage2_cli_agents_catalog.md	36KB	60+ CLI agents cataloged
stage2_multi_cli_comparison.md	60KB	10 agent comparison matrix
stage3_gap_analysis.md	64KB	137 features, 54 gaps identified
stage4_integration_plan.md	84KB	60 tasks, 6 phases, critical path
stage4_technical_documentation.md	519KB	29 features with Go code examples
stage4_testing_strategy.md	179KB	5,245+ tests across 8 categories
HELIX_MASTER_DOCUMENTATION.md	39KB	Executive summary + compiled master
Helix_Master_Documentation.docx	33KB	Word version of master document
integrate_submodules.sh	8KB	Automated SSH submodule integration script

Critical Stats

- **Repositories:** 9 analyzed (HelixCode + 8 submodules)
- **Submodules in HelixCode:** 87 declared, 4 missing, 4 empty
- **CLI Agents Cataloged:** 60+
- **Features Evaluated:** 137
- **Gaps to Close:** 54 (18 P0 critical)
- **Tests Required:** 5,245+
- **Estimated Effort:** 63 working days (~3 months, 3-4 engineers)